AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Real-Time Cyber-Physical Threat Detection for Industrial Control Systems

23CSE399 – Project Phase 1 (Panel Review 2)

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Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23003	Amarthya Sujai	**Dr. Anantha Narayanan V** *Associate Professor,* *Dept. of CSE*	**Type 1** Research	
CB.SC.U4CSE23020	Dwarakesh Venkatesh			
CB.SC.U4CSE23049	Shyam Sundar Raju			
CB.SC.U4CSE23050	Swetha C			
CB.SC.U4CSE23054	Sudharsana Saravanan S			

1 Introduction and Background

1.1 Context: ICS/SCADA in Critical Infrastructure

Industrial Control Systems (ICS) and Supervisory Control and Data Acquisition (SCADA) systems are centralized architectures that supervise, collect data, and provide operator control over distributed field devices. In modern critical infrastructure—particularly Water Treatment plants—these systems automatically control pumps, valves, and chemical dosing to ensure safe operations. However, IT/OT convergence has brought previously isolated Programmable Logic Controllers (PLCs) online. SCADA workstations and cloud dashboards now traverse shared enterprise networks.

1.2 Motivation & Challenges

Adversaries are evolving from noisy ransomware attacks to stealthy semantic/logic attacks (e.g., Stuxnet). These attacks subtly alter setpoints and control logic while the Human-Machine Interface (HMI) continues to display ”normal” operation. Water utilities urgently require mechanisms to detect these attacks without ”rip-and-replace” upgrades. Securing these environments presents distinct challenges:

- **Hardware Constraints:** Legacy PLCs offer limited CPU/memory, making heavy AI agents impractical.
- **Protocol Weaknesses:** Widely deployed Modbus TCP/IP provides no native authentication or integrity.
- **Latency Constraints:** Security controls must not disrupt sub-second control loops that protect physical assets.

1.3 Literature Survey

To understand existing defenses, the team analyzed 6 key research papers focusing on AI-based detection, statistical approaches, and Modbus protocol vulnerabilities.

Paper / Focus	Advantages	Limitations
Karacayılmaz et al. *AI-based IIoT* [1] (2024)	Adaptive to new threats; high accuracy using advanced AI models.	”Black-box” nature lacks explainability; high computational cost for legacy PLCs.
Murugesan et al. *Statistical Approach* [2] (2024)	Handles missing data effectively; mitigates early attack discrimination.	Limited to smart grids; not applicable to broader ICS environments.
Mohammed et al. *Field flooding* [3] (2023)	Specifically tailored for O&G infrastructure; effective against DoS.	Narrow scope (flooding only); may miss subtle Logic Injection attacks.
Barsha & Hubballi *State Transition*[4] (2024)	Detects semantic attacks by validating the *sequence* of operations.	Manual state modeling is complex; struggles with high-speed, concurrent states.
Athar et al. *Dual-Layer Attestation* [5] (2025)	Strong integrity verification; prevents firmware tampering.	High overhead for real-time loops; requires specific architecture changes.
Srivastava et al. *DDoS in SCADA* [6] (2024)	Critical protection against Availability attacks (DDoS) ensuring uptime.	Focuses on Network Layer (volumetric); misses Application Layer (logic) attacks.

2 Project Proposal and Objectives

2.1 Gap Formulation & Problem Statement

Research Gap: Existing systems either introduce unacceptable latency/resource overhead on legacy PLCs, or they focus only on network-level anomalies and fail to detect when a compromised PLC actively feeds physics-plausible but malicious values to the HMI.

Problem Statement: Design a lightweight, physics-aware intrusion detection system for an Industry 4.0 water treatment plant that can reliably catch stealthy semantic/logic attacks on legacy PLCs without modifying OT hardware or violating real-time control constraints.

2.2 Project Objectives

1. To simulate and analyze attacks in ICS networks using the Modbus TCP/IP protocol, including register manipulation and unauthorized command injection.
2. To design a lightweight monitoring architecture capable of observing PLC logic behavior and process data without introducing significant overhead to real-time control operations.
3. To validate the system's ability to detect physics-defying anomalies in real time by identifying inconsistencies between PLC-reported values and expected physical process behavior.

3 Execution Plan & Fundamentals Learned

3.1 Execution Plan (Subdomains & Approach)

To solve the identified problem, the team mapped the project into three core subdomains that needed to be mastered sequentially:

- **Subdomain 1: Industrial Protocol Mechanics.** Understanding the lack of encryption/authentication in Modbus TCP/IP to identify how threat actors inject commands.
- **Subdomain 2: Offensive OT Security.** Learning to safely execute DoS and Logic Manipulation attacks in a lab environment to generate a reliable dataset of compromised behavior.
- **Subdomain 3: State-Space & Physics Modeling.** Formulating mathematical models representing the physical constraints of a water treatment plant (e.g., flow rate limits, valve states).

3.2 Fundamentals Learned in the Semester

To build competency in the above subdomains, the team completed the following foundational studies:

- **Research Papers:** Extensively studied the 6 IEEE papers tabulated in Section 1 to understand state-of-the-art anomaly detection and its limitations.
- **Protocol Tutorials & Documentation:** Reviewed the official Modbus Application Protocol V1.1b3 specification to understand memory mapping (Coils, Discrete Inputs, Holding Registers).
- **Practical Python Tooling (MOOCs/Docs):** Completed hands-on tutorials for `pymodbus` and `scapy` to successfully interface with physical PLCs programmatically from an admin laptop.

4 Architecture Design and Explanation

To meet our objectives, we propose a two-tier architecture that validates system integrity by correlating internal controller logic with global physical process laws.

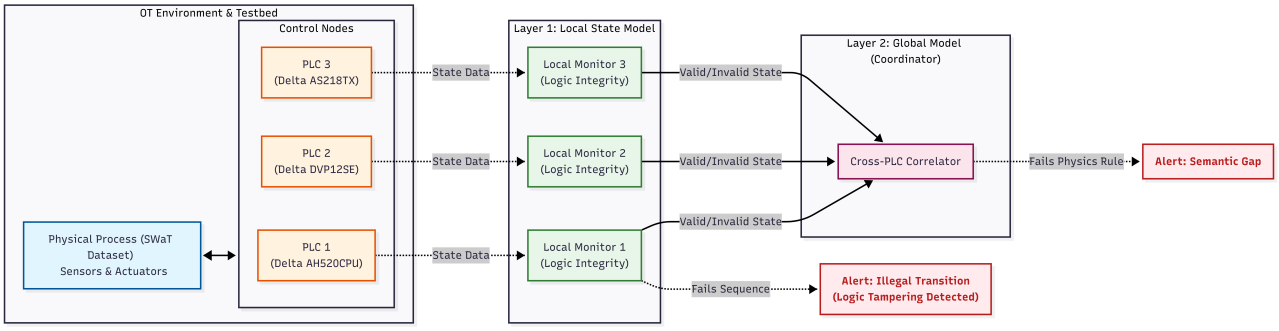


Figure 1: Hierarchical State-Based Monitoring Architecture

4.1 Explanation of Modules

- **Layer 1: Local State Model (PLC):** Focuses on Logic Integrity & Sequencing. This module monitors the internal transitions of individual PLCs. It detects "Illegal Transitions" (e.g., jumping from *Idle* to *High Pressure* without a *Ramping* state), effectively catching code-level tampering and unauthorized logic changes.

- **Layer 2: Global Model (Coordinator):** Focuses on Process Physics & Cross-PLC Correlation. It aggregates local state models to monitor the relationship between multiple assets (e.g., Pump vs. Tank). It detects "Semantic Gaps" where a PLC reports "Normal" operation, but the physical reality (flow/pressure) indicates a spike or drop.

5 Summary of Works Completed / Research Findings

5.1 ICS Testbed Implementation

We successfully constructed a functional ICS hardware testbed to serve as our physical validation ground. The control backbone utilizes Modbus TCP/IP (Port 502) and includes:

- **Control Nodes:** Delta AH520CPU (PLC 1), Delta DVP12SE (PLC 2), Delta AS218TX (PLC 3).
- **Network Infrastructure:** Edge Router (Delta DX-3021L9), Network Switch (TP Link TL-SG3021), and an Admin Laptop.

5.2 Attack Simulation & Technical Evidence

We practically validated the vulnerability of our testbed by executing a multi-stage attack kill chain mapped to the MITRE ATT&CK framework for ICS.

Kill Chain Phase	**Attack Action**	**MITRE ID**	**Execution & Result**
1. Reconnaissance	Network & Port Scanning	T0846, T0843	Mapped OT topology and discovered open Modbus (502) and Web (80) ports without triggering network alarms.
2. Intrusion	Session / Cookie Hijacking	T0859	Intercepted active authentication cookies from the Admin laptop to bypass IT login mechanisms entirely.
3. Execution	Register Manipulation (`pymodbus`)	T0821, T0836	Used Pymodbus to write to **Digital Coils** (forcing actuators) and **Analog Holding Registers** (altering pressure).
4. Impact	Denial of Service (DoS)	T0814	Flooded the PLC's network interface with high-frequency requests, exhausting CPU resources and halting the process.

6 Plans for the Oncoming Semesters

Having completed the literature review, testbed formulation, and vulnerability exploitation in Phase 1, our roadmap for the next semester focuses entirely on mitigation, evaluation, and delivery:

- **Phase 1: Dataset Baselining.** Simulate normal operational traffic and process physics from the Secure Water Treatment (SWaT) dataset onto our 3-PLC physical testbed to establish a physical ground truth.

- **Phase 2: Active Threat Injection.** Deploy our custom `pymodbus` exploitation framework against the active SWaT simulation to generate a controlled, compromised environment.

- **Phase 3: System Evaluation.** Deploy our proposed Hierarchical State-Based architecture. We will measure the system's accuracy in identifying semantic anomalies, track false-positive rates, and critically evaluate the computational latency added to the control loops.

- **Phase 4: Final Deliverables.** Publish the mathematical state-transition model, open-source the attack/defense scripts repository, and finalize the research thesis.

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